

Tagger Manual and Questionnaire

Literature Review:
Actionable XAI and Mechanistic Interpretability for LLMs

February 16, 2026

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1 General Notes

Explanation of tags:

Git

Also search for the answer to this question inside any found appendix (most likely a repo)

Trivial

Question that's very easy to find and can be answered by "dumb" model

Complicated

Question that likely requires deeper understanding of text and multiple passages

Automatic Extraction

Will be answered from semantic scholar data, bibtex entry, API etc. not by an LLM

Taxonomy

These are just suggestions to start with. During the pilot phase we will see how many entries and unclassifiables we get and then based on that merge entries or add new ones until we have a scheme of 3-8 categories (taxonomy development as per [5])

2 Abstract Screening

2.1 Language Availability

Trivial

Question: Is the paper in English?

Expected Values

Type: Binary

Values: YES — NO

2.2 Paper

Question: Is this a paper?

Expected Values

Type: Binary

Values: YES — NO

Anchor / Detailed Instructions

Answer with YES in case of doubt.

Exclude:

- Pure position papers / opinion pieces
- Lecture notes
- Tutorials

2.3 Review

Automatic Extraction

Question: Is the paper a review or a survey?

Expected Values

Type: Binary

Values: YES — NO

2.4 Model Type

2.5 XAI/MI Method Application

Editing Comment

Add to main process to prevent false positives

Question: Does the paper study a transformer-based language model?

Expected Values

Type: Binary

Values: YES — NO

Anchor / Detailed Instructions

Include:

- Decoder-only LLMs (GPT, LLaMA, Claude, etc.)
- Encoder-only models (BERT, RoBERTa) applied to language tasks
- Encoder-decoder models (T5, BART)
- Multimodal models with text component (LLaVA, GPT-4V, Gemini)
- Mixture-of-experts LLMs (Mixtral)

Exclude:

- Pure vision models (ViT without text)
- Diffusion models without text (image-only Stable Diffusion)
- Non-transformer architectures (RNNs, LSTMs, Mamba/SSMs as primary subject)
- Tabular/structured ML models

Edge cases:

- Mamba + Transformer hybrids: **YES** if transformer component is studied
- Papers comparing transformers to other architectures: **YES** if transformer results are substantive

2.6 XAI/MI Method Application

Editing Comment

Add to main process to prevent false positives

Question: Does the paper apply, propose, or evaluate a method to explain, understand, or modify internal model behavior using XAI or MI techniques?

Expected Values

Type: Binary

Values: YES — NO

Anchor / Detailed Instructions

Include papers that:

- Analyze internal representations (activations, attention patterns, hidden states)
- Use probing classifiers to understand what models encode
- Apply causal interventions (activation patching, causal tracing)
- Train or apply sparse autoencoders (SAEs) for feature analysis
- Use representation engineering / steering vectors
- Perform circuit-level analysis (identifying computational subgraphs)
- Apply gradient-based attribution methods to internal components

Exclude papers that:

- Only use LLMs as tools (e.g., LLM-as-judge, LLM for data labeling)
- Only conduct pure ablation studies
- Use LLMs to improve or work with XAI on non-LLM models
- Only evaluate model performance on benchmarks without interpretability
- Only propose adversarial attacks without analyzing *why* they work mechanistically

3 Metadata

For included papers, the below will extract important metadata using the first page and the abstract.

3.1 Metadata

3.1.1 Language

Trivial

Question: Is the full paper in English?

Expected Values

Type: Binary

Values: YES — NO

3.1.2 Affiliation

Automatic Extraction

Question: What are the affiliations of the primary authors

Expected Values

Type: Free Text

Separator: Semicolon

3.1.3 Venue

Automatic Extraction

Question: Where has the paper been published (what conference, which journal...)?

Expected Values

Type: Free Text

Separator: Semicolon

Example: NeurIPS; CauScien: Uncovering Causality in Science

3.2 OCR or other issues

Question: Are there indications that the text extraction process has produced serious errors that could impede tagging accuracy?

Expected Values

Type: Binary

Values: YES — NO

Anchor / Detailed Instructions

Yes

- Large parts of nonsensical characters
- Known artifacts of faulty OCR
- Sections being in wrong order

No

- Weird line breaks
- Wrong special characters

4 Core Extraction

4.1 Paper Classification

4.1.1 Role of XAI in Paper

Question: What primary role does XAI play in the paper?

Expected Values

Type: Single Select

Values:

- **XAI_AS_OBJECT:** XAI itself is the subject of study (e.g., method design, evaluation, critique, comparison).
- **XAI_AS_INSTRUMENT:** XAI is used as an analytical tool to generate insight about a model, system, or external domain.
- **INFRASTRUCTURE:** Tooling, datasets, benchmarks, or workflows for others.
- **HYBRID:** Multiple of the above

4.1.2 Target Domain

Question: What is the primary target domain of the insight?

Expected Values

Type: Multi Select

Values:

- **ML_MODEL_BEHAVIOR:** Internal representations, decisions, or failure modes of ML models.
- **SECURITY_SAFETY:** Safety and/or security aspects of either the model or of something using the model
- **DATASET_PROPERTIES:** Biases, artifacts, or structure in datasets.
- **NATURAL_SCIENCE:** Scientific domains such as biology, physics, or chemistry.
- **MEDICAL_OR_CLINICAL:** Medical, clinical, or healthcare-related analysis.
- **SOCIAL_OR_POLICY:** Social systems, governance, law, or policy.
- **HUMAN_FACTORS:** Human understanding, trust, usability, or decision-making.

4.1.3 Phenomenon Origin

Question: How does the studied phenomenon arise?

Expected Values

Type: Multi-select

Values:

- NATURALLY_OCCURRING: Emerges from standard behavior
- ADVERSARIALLY_INDUCED: Triggered by attacks
- TRAINING_INDUCED: Trained in
- FINE_TUNING_INDUCED: Emerges from fine-tuning
- NOT_APPLICABLE: Explain shortly in notes why

4.1.4 Object of Study

Question: What specific behavior, flaw, or capability is being studied?

Expected Values

Type: Multi-select

Values:

- HALLUCINATION_FACTUAL: Factual errors, confabulation.
- JAILBREAK_SAFETY: Circumventing safety guardrails/refusals.
- BIAS_FAIRNESS: Stereotyping, demographic representation.
- DECEPTION_SYCOPHANCY: Lying, sandbagging, or fawning.
- MEMORIZATION_PRIVACY: Training data extraction.
- DANGEROUS_CONTENT: CBRN, cyber-offense info (distinct from jailbreak mechanism).
- GENERAL_REASONING: How the model does math, logic, or IOI (Indirect Object Identification).
- INTERNAL_MECHANISM: Studying a circuit/feature without a specific behavioral flaw attached (e.g., “polysemanticity” in the abstract).
- OTHER: Specify in notes.

4.1.5 XAI Method

Taxonomy

Question: What XAI method does the paper use or look at?

Expected Values

Type: Multi-select

Values:

- **FEATURE_DECOMPOSITION:** Sparse Autoencoders (SAEs), PCA, Dictionary Learning.
- **CAUSAL_INTERVENTION:** Activation patching, causal tracing, ablation (zeroing/mean-ablating).
- **PROBING:** Training classifiers (Linear or MLP) on activations.
- **ATTENTION_ANALYSIS:** Analyzing attention heads/patterns directly.
- **REPRESENTATION_STEERING:** Steering vectors, activation addition/subtraction.
- **GRADIENT_ATTRIBUTION:** Saliency maps, integrated gradients.
- **VOCAB_PROJECTION:** Logit lens, tuned lens.
- **CIRCUIT_DISCOVERY:** Identifying specific subgraphs/computational paths.
- **WEIGHT_EDITING:** ROME, MEMIT (permanent parameter changes).
- **EXAMPLE_BASED:** Counterfactuals or Influential training examples
- **NATURAL_LANGUAGE:** Model generates a textual explanation/rationale
- **OTHER:** Anything not fitting into the above

4.1.6 Model Access Required

Question: What minimum access level does the method require?

Expected Values

Type: Single-select (Minimum needed)

Values:

- **BLACK_BOX:** API input/output only
- **PARTIAL_LOGITS:** Token probabilities but only for top X tokens
- **LOGITS:** Token probabilities
- **ACTIVATIONS:** Internal representations
- **WEIGHTS:** Full static weights
- **GRADIENTS:** Gradient access
- **TRAINING_DATA:** Training-time access
- **NOT_APPLICABLE:** Shortly explain why N/A in notes

4.1.7 Lifecycle Stage

Question: When is the method applied?

Expected Values

Type: Multi-select

Values:

- **PRE_TRAINING:** Applied to the base model training loop.
- **POST_TRAINING:** Applied during SFT/RLHF.
- **INFERENCE:** Applied live during generation (e.g., steering).
- **POST_HOC:** Applied after generation is complete (analysis).
- **OTHER:** Specify in notes

4.2 Models and Scale

4.2.1 Model Families

Question: Which specific models are evaluated?

Expected Values

Type: Free Text (Semicolon-separated).

Example: GPT-4; LLaMA-2-7B-Chat; Pythia-70M

Anchor / Detailed Instructions

Write this down exactly as mentioned in the paper. This includes all fine tuned versions, mentioned parameter counts etc

Editing Comment

I think it may be best to simply embrace the free form nightmare here and later cluster

5 Reproducibility Assessment

5.1 Code Availability

5.1.1 Implementation Access

Question: Is source code for the method/experiments available? [2, 1]

Expected Values

Type: Single-select

Values:

- `PUBLIC_WITH_URL` (Direct link provided in paper)
- `STATED_UPON_REQUEST` (Authors promise to share)
- `NOT_AVAILABLE` (No code mentioned or explicitly not shared)
- `PSEUDOCODE_ONLY`: Only algorithmic logic described, no executable code.
- `NOT_APPLICABLE` (Pure theoretical work)

5.1.2 Repository URLs

Question: What is (are) the repository URL(s)?

Expected Values

Type: Free Text or NA. Separate with semicolon if multiple.

5.1.3 Pre-processing and Pipeline Code

Question: Is the code for data pre-processing (tokenization, filtering, prompting wrappers) included? [1]

Expected Values

Type: Single-select

Values:

- `INCLUDED_IN_REPO` (Part of the main codebase)
- `SEPARATE_APPENDIX` (Provided in appendix/supplement)
- `DESCRIBED_TEXT_ONLY` (Text description but no script)
- `MISSING` (Not provided)
- `NOT_APPLICABLE`

5.1.4 Code Documentation Quality

Question: What level of documentation exists for the code?

Expected Values

Type: Multi-select

Values:

- README_EXISTS (Any README file present)
- INSTALLATION_INSTRUCTIONS (How to set up environment)
- USAGE_EXAMPLES (Example commands or notebooks)
- INLINE_COMMENTS (Code has meaningful comments)
- API_DOCUMENTATION (Generated docs or docstrings)
- NONE_FOUND

5.1.5 License

Question: Is the code released under an explicit license? Automatic Extraction

Expected Values

Type: Free Text or MISSING

5.2 Practitioner Usability and Popularity

5.2.1 Package Distribution

Taxonomy Automatic Extraction

Question: How is the code distributed?

Expected Values

Type: Multi-select

Values:

- GIT_REPO
- PYPI_PACKAGE (Available via `pip install`)
- CONDA_PACKAGE (Available via `conda`)
- HUGGINGFACE_INTEGRATION (Integrated into HuggingFace libraries)
- OTHER_PACKAGE_MANAGER (e.g., npm, cargo)
- DOCKER_CONTAINER
- NOT_APPLICABLE

5.2.2 Maintenance Activity Indicators

Automatic Extraction

Question: What indicators of ongoing maintenance exist?

Expected Values

Type: Multi-select

Values:

- RECENT_COMMITS (Commits within 6 months of tagging date)
- ACTIVE_ISSUE_RESPONSES (Issues/PRs addressed by maintainers)
- MULTIPLE_CONTRIBUTORS (More than original authors contributed)
- VERSIONED_RELEASES (Tagged releases / semantic versioning)
- STALE (No activity for > 1 year)
- ARCHIVED (Explicitly archived or deprecated)
- NOT_ASSESSABLE (Repository statistics unavailable)

5.2.3 Adoption Metrics

Automatic Extraction

Question: What quantitative indicators of adoption exist?

Expected Values

Type: Free Text (record actual numbers) or NOT_STATED

Fields:

- GitHub stars (if applicable):
- Forks:
- PyPI / package downloads (monthly, if stated):

5.2.4 Post-Publication Maintenance

Automatic Extraction

Question: What is the post-publication maintenance status?

Expected Values

Type: Free Text or NOT_AVAILABLE

Fields:

- Date of last commit:
- Total number of commits:

5.3 Data Availability

5.3.1 Dataset Access

Taxonomy Automatic Extraction

Question: Are the datasets used in experiments available?

Expected Values

Type: Single-select

Values:

- FULLY_PUBLIC (All datasets are publicly available benchmarks or released by authors)
- PARTIALLY_PUBLIC (Some datasets public, some private)
- NEW_PUBLIC: Introduces a new dataset/subset and provides a link/download.
- PROPRIETARY_PRIVATE: Uses internal company data or unreleased user data.
- GENERATED_REPRODUCIBLE: Data is synthetic (LLM-generated) *and* the prompts/seeds to regenerate it are provided.
- GENERATED_OPAQUE: Data is synthetic but prompts/methods to regenerate are missing.
- NO_DATASET_USED (Theoretical or no empirical evaluation)

5.3.2 Datasets Used

Question: Which datasets are used?

Expected Values

Type: Free Text separate by semicolons

Values: All names explicitly referenced. NONE if datasets are not explicitly named/referenced.

5.3.3 Data Splits and Raw Artifacts

Question: Are specific data splits and intermediate results shared? [2]

Expected Values

Type: Multi-select

Values:

- SPLITS_SHARED (Explicit Training/Validation/Test sets provided/defined)
- INTERMEDIATE_RESULTS (Raw model outputs, logits, or activations shared)
- NONE

Anchor / Detailed Instructions

[2] explicitly requires "Validation data", "Test data", and "Results" (intermediate outputs) to be shared, not just the raw input data. I assume 90%+ of papers will not meet requirement but we can use this in analysis.

5.4 Computational Reproducibility

5.4.1 Compute Resources Reporting

Question: How detailed is the reporting of computational requirements? [2, 1]

Expected Values

Type: Multi-select

Values:

- `HARDWARE_SPECIFIED` (GPU/CPU model named)
- `TIME_REPORTED` (Wall-clock time or GPU hours)
- `COST_REPORTED` (Monetary cost or compute budget)
- `MEMORY_SPECIFIED` (RAM/VRAM requirements)
- `SOFTWARE_ENVIRONMENT` (OS, CUDA version, framework versions)
- `NOT_REPORTED`

5.4.2 Exact Compute Resources Reporting

Question: If given, what is specifically mentioned for computational requirements?

Expected Values

Type: Free-text

NONE or the full info that is mentioned

5.4.3 Randomness and Seeds

Question: Is the handling of randomness described?[1]

Expected Values

Type: Multi-select

Values:

- `SEEDS_SPECIFIED` (Exact random seeds provided)
- `SEEDING_PROCESS_DESCRIBED` (How seeds are set)
- `MULTIPLE_RUNS_REPORTED` (Results from multiple seeds)
- `NOT_ADDRESSED`
- `NO_RANDOMNESS`

5.5 Methodological Transparency

5.5.1 Method Description Completeness

Question: How thoroughly is the method described?

Expected Values

Type: Single-select

Values:

- `ALGORITHMIC` (Pseudocode or mathematical formulation provided)
- `DETAILED_PROSE` (Step-by-step description without formal algorithm)
- `HIGH_LEVEL_ONLY` (Conceptual description, missing implementation details)
- `REFERENCE_ONLY` (Points to another paper without restating)

5.5.2 Hyperparameter Reporting

Question: Are hyperparameters documented?[2, 1]

Expected Values

Type: Multi-select

Values:

- `FINAL_VALUES_LISTED` (All hyperparameters for reported results given)
- `SEARCH_SPACE_DESCRIBED` (Range of values tried during tuning)
- `SELECTION_CRITERION_STATED` (How final values were chosen)
- `PARTIAL_REPORTING` (Some but not all parameters given)
- `NOT_REPORTED`
- `NO_APPLICABLE`

6 Evidence Assessment

6.1 Experimental Design

6.1.1 Scientific Structure

Question: Does the paper explicitly state the scientific components? [2]

Expected Values

Type: Multi-select

Values:

- EXPLICIT_PROBLEM (Clear mention of the problem to solve)
- EXPLICIT_OBJECTIVE (Clear mention of research objective)
- EXPLICIT_HYPOTHESIS (Formal hypothesis stated *before* experiments)
- EXPLICIT_PREDICTION (Predictions related to hypothesis)
- NONE_IMPLICIT_ONLY

6.1.2 Does Paper Include Empirical Evaluation?

Question: Does the paper present experimental results or empirical analysis?

Expected Values

Type: Single-select

Values:

- YES_QUANTITATIVE (Numerical results, metrics, benchmarks)
- YES_QUALITATIVE (Case studies, visualizations only)
- YES_MIXED (Both quantitative and qualitative)
- NO_PURELY_THEORETICAL (No experiments)

6.1.3 Distinction of Speculation

Question: Does the paper clearly delineate between objective results and speculation? [1]

Expected Values

Type: Single-select

Values:

- CLEAR_DELINEATION (Speculation/Hypothesis clearly marked vs Facts)
- BLURRED (Narrative blends results and opinion)
- FACTS_ONLY (Minimal interpretation provided)

Editing Comment

Probably wont work but idea is that in 6.1.1. you could write down what it is and compare to conclusion. E.g. if paper claims that A is better than B but did not give any form of significance testing, errors etc it can be counted as blurred.

6.1.4 Evaluation Dataset Scale

Taxonomy

Question: What is the scale of the evaluation dataset(s)?

Expected Values

Type: Single-select (select the largest applicable)

Values:

- ANECDOTAL (< 10 examples, hand-picked)
- SMALL (10 – 99 examples)
- MEDIUM (100 – 999 examples)
- LARGE (1000 – 9999 examples)
- VERY_LARGE (≥ 10000 examples)
- STANDARD_BENCHMARK (Named benchmark like MMLU, GSM8K)
- NOT_STATED
- NOT_APPLICABLE

Follow-up: If exact number stated, record it in notes.

Anchor / Detailed Instructions

If multiple, record the biggest mentioned number

Editing Comment

This needs to be specified further. There are papers that use standard benchmarks but only use parts of them. Also this is too granular

6.2 Statistical Rigor

6.2.1 Uncertainty Quantification

Question: How is uncertainty in results reported? [1]

Expected Values

Type: Multi-select

Values:

- VARIANCE_REPORTED: Error bars, standard deviations, or confidence intervals.
- SIGNIFICANCE_TESTING: P-values or hypothesis tests (e.g., t-test, Wilcoxon).
- MULTIPLE_SEEDS: Results explicitly stated to be averaged over more than one random seed.
- NONE: Only point estimates (averages or single runs) reported.

6.2.2 Number of Experimental Runs

Question: How many times were experiments run? [1]

Expected Values

Type: Single-select

Values:

- SINGLE_RUN
- MULTIPLE_STATED (Explicit: "averaged over N runs")
- NOT_STATED

Follow-up: If number stated, record it in notes.

Editing Comment

This could maybe be absorbed into Uncertainty Quantification i.e. add an option there for Only Single Run

6.3 Baselines and Comparisons

6.3.1 Comparison Strategy

Question: What are results compared against?

Expected Values

Type: Multi-select

Values:

- PRIOR_METHODS (Published competing methods)
- ABLATIONS (Removing components of proposed method)
- NAIVE_BASELINE (Random, zero-shot, or trivial baseline)
- HUMAN_PERFORMANCE (Human annotations or judgments)
- NO_COMPARISON (Standalone results only)

6.3.2 Justification of Dataset Choice

Question: Is there explicit justification for why these datasets/benchmarks were chosen?

Expected Values

Type: Single-select

Values:

- EXPLICIT_JUSTIFICATION (Clear explanation of dataset choice)
- NO_JUSTIFICATION

6.4 Evaluation Metrics

6.4.1 Metrics Used

Question: What evaluation metrics are employed?

Expected Values

Type: Multi-select

Values:

- STANDARD_ML_METRICS (Accuracy, F1, AUROC, etc.)
- DOMAIN_SPECIFIC_METRICS (Custom metrics for the studied phenomenon)
- HUMAN_EVALUATION (Human ratings or annotations)
- LLM_AS_JUDGE (LLM evaluates quality)
- QUALITATIVE_ONLY (Visual inspection, no quantitative metric)
- NOT_APPLICABLE

6.4.2 Metric Justification

Question: Are the chosen metrics justified or defined? [1] Taxonomy

Expected Values

Type: Single-select

Values:

- `FORMALLY_DEFINED` (Mathematical definition or citation provided)
- `BRIEF_EXPLANATION` (Informal description)
- `MOTIVATED_TEXT` (Explanation of why metric was chosen)
- `NOT_EXPLAINED`

6.5 XAI Specifics

6.5.1 Explanation Scope

Question: What is the scope of the explanation provided? [4]

Expected Values

Type: Single-select

Values:

- LOCAL (Explains individual predictions/instances)
- GLOBAL (Explains overall model behavior/logic)
- CONCEPT_BASED (Explains via high-level human concepts)
- HYBRID (Combines local and global approaches)

6.5.2 Evaluation Paradigm

Question: How is the explanation method evaluated? [4]

Expected Values

Type: Multi-select

Values:

- ANECDOTAL_ONLY (Visual examples chosen by authors, no metrics)
- FUNCTIONAL_QUANTITATIVE (Proxy metrics: Faithfulness, Comprehensibility, Sufficiency)
- HUMAN_USER_STUDY (Experiments with real human participants)
- APPLICATION_GROUNDED (Evaluated on downstream task performance improvement)
- LLM_RATING (LLM judges if explanation makes sense)
- NOT_ASSESSED
- NOT_APPLICABLE

6.5.3 Human Evaluation Type

Question: If a user study was conducted, what was measured? [3]

Expected Values

Type: Multi-select

Values:

- NOT_APPLICABLE (No user study)
- SUBJECTIVE_RATING (Likert scales for trust, satisfaction, clarity)
- SIMULATABILITY (Can humans predict model output using the explanation?)
- TASK_PERFORMANCE (Does the explanation help humans perform a task faster/better?)
- DEBUGGING (Can humans identify model errors using the explanation?)

Editing Comment

sample size?

6.5.4 Faithfulness Verification

Question: Is the causal relevance of identified components verified?

Expected Values

Type: Multi-select

Values:

- ABLATION_BREAKS_BEHAVIOR (Removing component destroys capability)
- RESTORATION_TEST (Adding component back restores behavior)
- QUANTITATIVE_ATTRIBUTION (Percentage of behavior explained measured)
- CORRELATION_ONLY (Component activates when behavior occurs, no causal test)
- NOT_VERIFIED
- NOT_APPLICABLE

Editing Comment

can probably unify ablation and restoration into one

6.6 Limitations and Failure Analysis

6.6.1 Failure Cases

Question: Are failure modes or limitations discussed?

Expected Values

Type: Single-select

Values:

- SYSTEMATIC_FAILURE_ANALYSIS (Dedicated analysis of when/why method fails)
- LIMITATIONS_SECTION (Standard limitations section with substantive content)
- BRIEF_ACKNOWLEDGMENT (Cursory mention of limitations)
- NONE_DISCUSSED (Only positive results presented)

6.6.2 Negative / Falsification Efforts

Question: Does the paper present evidence that contradicts, disproves, or fails to support a claim, hypothesis, or method?

Expected Values

Type: Single-select

Values:

- DISPROVES_PRIOR_CLAIM (Shows existing XAI method/claim doesn't work as stated)
- NULL_RESULT_OWN_HYPOTHESIS (Authors' hypothesis not supported by their data)
- NEGATIVE_COMPARISON (Proposed method performs worse than baselines/prior work)
- NO_SIGNAL_FOUND (Attempted analysis but found no meaningful pattern/effect)
- NONE (Only positive or confirmatory results)

References

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